



DS&AI LAB PROJECT | TERM MAY 2026

Course-Aware Personalized Learning Companion for IIT Madras BS Students

Milestone 1 : Problem Definition & Literature Review

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The Problem & Project Objectives

Scattered course resources, and how we plan to unify them



The Problem

- Lectures, notes, slides, PYQs, FAQs and forums live on separate platforms - none aware of the others.
- Revising one concept means searching a video, a PDF and a forum thread separately.
- Existing AI tools don't stay aware of a specific course, remember a student's weak spots, or link doubts to exact lecture moments.

Primary Objective

A prototype learning companion that unifies one course's resources into a source-grounded knowledge base and personalizes learning to each student.



Source-grounded RAG Q&A



Lecture timestamp navigation



Persistent learner profile & gap detection



AI Question Intelligence (PYQ/FAQ dedup)

Literature Review: Where Existing Tools Fall Short

Three tool categories reviewed - each strong in isolation, none course-aware end-to-end



NotebookLM

Document-grounded AI assistant

STRENGTH

Source-grounded answers with citations; summaries, mind maps, quizzes

GAP FOR THIS PROJECT

No standing course model; no persistent learner profile; no timestamp linking



Adaptive Quiz Tools

AI-based assessment platforms

STRENGTH

Adjusts difficulty; tracks performance within its own environment

GAP FOR THIS PROJECT

Not built for open-ended, source-grounded Q&A; manual question banks



General LLM Assistants

Broad conversational AI

STRENGTH

Flexible, answers almost anything in natural language instantly

GAP FOR THIS PROJECT

Not grounded in course materials; hallucination risk; no memory of student progress



Key gap identified: no single tool combines source-grounded, course-specific answering with a persistent learner model and a self-maintaining question repository - this is the space our project targets.

Progress So Far & Team Contributions

Milestone 1 complete - moving next into system design and data collection

Progress Snapshot

- ✔ Problem statement finalized & scoped to one BS course
- ✔ Literature review of NotebookLM, quiz tools & LLM assistants
- ✔ Gap analysis & opportunity mapping completed
- ⋮ Next: dataset collection & RAG pipeline design

Individual Contributions - Milestone 1

Member	Contribution
Mayank Singh	Problem definition, scoping & objectives drafting
Ali Jawad	NotebookLM research & source-grounded RAG analysis
Sachi Dhaturaha	Adaptive quiz/assessment tools review & comparison table
Aaryan Pratap Maurya	General LLM assistant review & gap analysis
Jibin V Mathews	Opportunities, conclusion & report compilation

All members contributed to review, discussion and finalization of this milestone report.